Mensuration

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(5)$.

Final Answer: 21. Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for percentages with technology consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for finding the whole (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 6x + 9$ in percentage increase and decrease, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for percentage increase and decrease consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

£800 is invested at 3.5% compound interest per year. Find the amount after 4 years, to the nearest penny.

$A = 800\left(1 + \frac{3.5}{100}\right)^4 = 800 \times 1.035^4 \approx 800 \times 1.1475 = £918.00$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the magnitude of $\mathbf{v} = \begin{pmatrix} -5 \\ 12 \end{pmatrix}$ and hence find a unit vector in the direction of $\mathbf{v}$.

$|\mathbf{v}| = \sqrt{25 + 144} = \sqrt{169} = 13$. Unit vector $= \frac{1}{13}\begin{pmatrix} -5 \\ 12 \end{pmatrix}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A tower PQ stands vertically. From point A on level ground, 50 m from the base Q, the angle of elevation of P is 32°. Find the height of the tower.

$h = 50\tan 32° \approx 50 \times 0.6249 \approx 31.2$ m.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

Total sum of the 5 scores = $5 \times 84 = 420$. Sum of the 4 known scores = $78 + 86 + 92 + 80 = 336$. 5th score = $420 - 336 = 84$. Removing the lowest score (78) leaves four scores: 80, 84, 86, 92. New sum = $420 - 78 = 342$. New mean = $342/4 = 85.5$ (an increase of 1.5 points).

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD $6cmandtheradiusisUSD3$ cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A block of mass 5 kg is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is 8 m and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8$ m/s^2, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4$ m. Initial gravitational potential energy: PE $= mgh = 5(9.8)(4) = 196$ J 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}$ N. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}$ N. Work done against friction over distance $d = 8$ m: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896$ J 3. By.

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

(a) Use the cosine rule in triangle OAB with $OA = OB = 10$ and $\angle AOB = \frac{2\pi}{3} = 120^\circ$: $AB^2 = 10^2 + 10^2 - 2(10)(10)\cos\left(\frac{2\pi}{3}\right) = 100 + 100 - 200\left(-\frac{1}{2}\right) = 200 + 100 = 300$ Thus $AB = \sqrt{300} = 10\sqrt{3}$ cm. (b) Area of sector OAB: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(10^2)\left(\frac{2\pi}{3}\right) = \frac{100\pi}{3}$ cm^2 Area of triangle OAB.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Number**

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

(a) At maximum height, $v = 0$. $u = +15$ m/s, $a = -9.8$ m/s^2: $v^2 = u^2 + 2as \Rightarrow 0 = 15^2 + 2(-9.8)s \Rightarrow 19.6s = 225 \Rightarrow s = \frac{225}{19.6} \approx 11.48$ m ≈ 11.5 m (b) At the sea, the displacement relative to the launch point is $s = -40$ m. Use $s = ut + \frac{1}{2}at^2$: $-40 = 15t + \frac{1}{2}(-9.8)t^2 \Rightarrow -40 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 40 = 0$ Solve via quadratic formula.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 10.01 Sets & Venn Diagrams**

For what value of k will the system have no solution?

$\{3x - 2y = 8;\ 6x + ky = 5$ Solution: 1. A linear system has no solution if the lines are parallel (identical slopes, different y-intercepts). 2. Express the first line in slope-intercept form: $-2y = -3x + 8 \Rightarrow y = \frac{3}{2}x - 4 \quad \Rightarrow m_1 = \frac{3}{2}$. 3. Express the second line in slope-intercept form: $ky = -6x + 5 \Rightarrow y = -\frac{6}{k}x + \frac{5}{k} \quad \Rightarrow m_2 = -\frac{6}{k}$. 4. Equate slopes to make lines parallel.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Algebra & Graphs**

A small particle of weight 24 N is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

Notice that the angle between string A and string B is $30^\circ + 60^\circ = 90^\circ$, so the strings are perpendicular. Resolve horizontally: $T_A\sin 30^\circ = T_B\sin 60^\circ \Rightarrow T_A(0.5) = T_B\left(\frac{\sqrt{3}}{2}\right) \Rightarrow T_A = \sqrt{3}T_B$ Resolve vertically: $T_A\cos 30^\circ + T_B\cos 60^\circ = 24 \Rightarrow T_A\left(\frac{\sqrt{3}}{2}\right) + T_B(0.5) = 24$ Substitute $T_A = \sqrt{3}T_B$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

Factor 2 from the first two terms: $2\left(x^2 - \frac{5}{2}x\right) - 3 = 0$. Halve the coefficient of x: $\frac{1}{2}\left(-\frac{5}{2}\right) = -\frac{5}{4}$. Complete the square: $2\left(x - \frac{5}{4}\right)^2 - 2\left(\frac{25}{16}\right) - 3 = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8} - \frac{24}{8} = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 = \frac{49}{8}$. Divide by 2: $\left(x - \frac{5}{4}\right)^2 = \frac{49}{16}$. Take square roots.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry

A cone and a cylinder have identical base radii of 6 cm. If both solids have the same volume of 180π cm^3, find the height of the cylinder and the height of the cone.

Solution: The base area for both solids is $B = \pi r^2 = \pi(6^2) = 36\pi$ cm^2. For the cylinder: $V_{\text{cyl}} = B \cdot h_{\text{cyl}} \Rightarrow 180\pi = 36\pi \cdot h_{\text{cyl}} \Rightarrow h_{\text{cyl}} = \frac{180\pi}{36\pi} = 5$ cm. For the cone:

$V_{\text{cone}} = \frac{1}{3}B \cdot h_{\text{cone}} \Rightarrow 180\pi = \frac{1}{3}(36\pi) \cdot h_{\text{cone}} = 12\pi \cdot h_{\text{cone}} \Rightarrow h_{\text{cone}} = \frac{180\pi}{12\pi} = 15$ cm. .

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

(a) Complete the squares for x and y: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 \Rightarrow (x - 3)^2 + (y + 2)^2 = 25$ The centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$. (b) The gradient of the radius from centre $(3, -2)$ to $P(6, 2)$ is: $m_{\text{rad}} = \frac{2 - (-2)}{6 - 3} = \frac{4}{3}$ The tangent is perpendicular to the radius, so its gradient is $m_{\tan} = -\frac{3}{4}$. Using point-slope form with $P(6, 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(2)$.

a. 16 ✓

b. 22

c. 12

d. 20

Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for converting between percentages, fractions, and decimals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for quantities as percentages consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arc of a circle of radius 9 cm subtends an angle of 120° at the centre. Find the arc length in terms of π.

a. 6π cm ✓

b. 3π cm

c. 9π cm

d. 12π cm

Arc length $= \frac{120}{360} \times 2\pi(9) = \frac{1}{3} \times 18\pi = 6\pi$ cm. [M1 A1]

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A cone has radius 9 cm and height 6 cm. Find its volume in terms of π.

a. 162π **cm**^{3} ✓

b. 152π cm^3

c. 174π cm^3

d. 159π cm^3

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 9^2 \times 6 = 162\pi$ cm^3.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A closed cylinder has radius 2 cm and height 2 cm. Find its total surface area in terms of π.

a. 16π **cm**^{2} ✓

b. 25π cm^2

c. 27π cm^2

d. 28π cm^2

SA $= 2\pi r^2 + 2\pi rh = 2\pi \times 2 \times (2 + 2) = 16\pi$ cm^2.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cylinder has radius 2 cm and height 10 cm. Find its volume in terms of π.

a. 50π cm^3

b. 29π cm^3

c. 40π **cm**^{3} ✓

d. 46π cm^3

Volume $= \pi r^2 h = \pi \times 2^2 \times 10 = 40\pi$ cm^3.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A cuboid has a length of 8 cm, a width of 5 cm, and a height of 3 cm. Calculate the volume of the cuboid.

a. **120 cm**^{3} ✓

b. 68 cm^3

c. 40 cm^3

d. 128 cm^3

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = 8 cm, width = 5 cm, height = 3 cm. Volume = 8 cm × 5 cm × 3 cm Volume = 40 cm^2 × 3 cm Volume = 120 cm^3. Therefore, the volume of the cuboid is 120 cm^3.

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30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A frustum of a right cone has top radius 3 cm, bottom radius 6 cm, and vertical height 4 cm. What is the volume of this frustum in terms of π?

a. 84π cm^3 ✓

b. 76π cm^3

c. 96π cm^3

d. 108π cm^3

Formula for volume of frustum: $V = \frac{1}{3}\pi h(R^2 + Rr + r^2) = \frac{1}{3}\pi(4)(6^2 + 6(3) + 3^2) = \frac{4}{3}\pi(36 + 18 + 9) = \frac{4}{3}\pi(63) = 4 \times 21\pi = 84\pi$ cm^3. [M1 for frustum volume formula or similar cones subtraction, A1 for 84pi]

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In a circle of radius 12 cm, a chord subtends a central angle of 90°. Calculate the area of the minor segment. Give your answer to 1 decimal place.

a. **41.1 cm²** ✓

b. 36.0 cm²

c. 45.2 cm²

d. 52.8 cm²

Area of sector = $\frac{90}{360}\pi(12^2) = \frac{1}{4}(144\pi) = 36\pi \approx 113.10$ cm². Area of right-angled triangle = $\frac{1}{2}(12)(12) = 72$ cm². Area of segment = $113.10 - 72 = 41.1$ cm². [M1 for sector area, M1 for triangle area, A1 for 41.1]

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Calculate the total surface area of a solid hemisphere with radius 5 cm. Give your answer in terms of π.

a. 50π cm^2

b. 75π cm^2 ✓

c. 100π cm^2

d. 25π cm^2

Total surface area of a solid hemisphere = curved surface area + circular base = $2\pi r^2 + \pi r^2 = 3\pi r^2 = 3\pi(5^2) = 75\pi$ cm^2. [M1 for 3*pi*r^2, A1 for 75pi]

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Calculate the total surface area of a cone with radius 7 cm and slant height 25 cm. Give your answer in terms of π.

a. 175π cm^2

b. 224π cm^2 ✓

c. 245π cm^2

d. 300π cm^2

Total surface area = curved surface area + base area = $\pi rl + \pi r^2 = \pi(7)(25) + \pi(7^2) = 175\pi + 49\pi = 224\pi$ cm^2. [M1 for pi*r*l + pi*r^2, A1 for 224pi]

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A copper wire of diameter 2 mm has a total volume of 50π cm^3. What is the length of the wire in metres?

a. **50 m** ✓

b. 5 m

c. 500 m

d. 25 m

Radius = 1 mm = 0.1 cm. Cross-sectional area = $\pi(0.1)^2 = 0.01\pi$ cm^2. Length = $\frac{50\pi}{0.01\pi} = 5000$ cm $= 50$ m. [M1 A1]

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A right pyramid has a square base of side 6 cm and a vertical height of 10 cm. What is its volume?

a. **120 cm³** ✓

b. 360 cm³

c. 60 cm³

d. 180 cm³

Base area = $6 \times 6 = 36$ cm^2. Volume = $\frac{1}{3} \times$ Base $\times\, h = \frac{1}{3} \times 36 \times 10 = 120$ cm^3. [M1 A1]

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 6 cm. Calculate the area of a sector with central angle 60°. Give your answer in terms of π.

a. 6π cm^2 ✓

b. 12π cm^2

c. 36π cm^2

d. 2π cm^2

Sector area: $A = \frac{\theta}{360^\circ}\pi r^2 = \frac{60}{360}\pi(6^2) = \frac{1}{6}\pi(36) = 6\pi$ cm^2. [M1 for formula, A1 for 6pi]

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A solid hemisphere has radius 4 cm. What is its TOTAL surface area (including the flat circular base) in terms of π?

a. 48π cm^2 ✓

b. 32π cm^2

c. 64π cm^2

d. 16π cm^2

Total SA of a solid hemisphere = curved hemisphere ($2\pi r^2$) + circular base (πr^2) = $3\pi r^2 = 3\pi(4)^2 = 48\pi$ cm^2. [M1 A1]

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry

A sector of a circle has central angle 40° and area 16π cm^2. Find the radius of the circle.

a. **12 cm** ✓

b. 6 cm

c. 8 cm

d. 16 cm

Sector area = $\frac{40}{360} \times \pi r^2 = 16\pi \Rightarrow \frac{1}{9}r^2 = 16 \Rightarrow r^2 = 144 \Rightarrow r = 12$ cm. [M1 A1]

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry

A sphere has radius 3 cm. Calculate its volume in terms of π.

a. 12π cm^3

b. 27π cm^3

c. 36π cm^3 ✓

d. 108π cm^3

Volume of a sphere: $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3^3) = \frac{4}{3}\pi(27) = 36\pi$ cm^3. [M1 for formula, A1 for 36pi]

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

A cone has radius 3 cm and vertical height 4 cm. What is its curved surface area in terms of π?

a. 15π cm^2 ✓

b. 12π cm^2

c. 30π cm^2

d. 36π cm^2

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{3^2 + 4^2} = 5$ cm. Curved SA = $\pi r l = \pi(3)(5) = 15\pi$ cm^2. [M1 A1]

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Transformations & Vectors

Two similar solid cylinders have heights in the ratio 2 : 3. If the total surface area of the smaller cylinder is 40 cm², what is the total surface area of the larger cylinder?

a. 90 cm² ✓

b. 60 cm²

c. 80 cm²

d. 135 cm²

Area scale factor = $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$. Larger area = $40 \times \frac{9}{4} = 90$ cm^2. [M1 A1]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Probability & Statistics

Calculate the total perimeter of a sector with radius 7 cm and angle 60° (to 1 decimal place).

a. 21.3 cm ✓

b. 26.3 cm

c. 7.3 cm

d. 14 cm

Perimeter of sector = $2r + \text{arc length} = 2(7) + \frac{60}{360}(2\pi \times 7) = 14 + 7.3 = 21.3$ cm. [M1 A1]

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Functions

A solid cone has radius 5 cm and slant height 13 cm. Calculate its vertical height.

a. 10 cm

b. 11 cm

c. 12 cm ✓

d. 14 cm

By Pythagoras: $h = \sqrt{l^2 - r^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm. [M1 for Pythagoras, A1 for 12]

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number

The curved surface area of a cylinder with radius 5 cm is 70π cm^2. What is the height of the cylinder?

a. 7 cm ✓

b. 14 cm

c. 5 cm

d. 10 cm

Curved SA = $2\pi rh = 70\pi \Rightarrow 2\pi(5)h = 70\pi \Rightarrow 10h = 70 \Rightarrow h = 7$ cm. [M1 A1]

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A rectangular swimming pool is 25 m long, 10 m wide, and has a uniform depth of 2 m. How many cubic metres of water does it hold when completely full?

a. **500 m³** ✓

b. 250 m³

c. 1000 m³

d. 750 m³

Volume = length × width × depth = 25 × 10 × 2 = 500 m^3. [B1]

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